

Imperial College London

BSc/MSci EXAMINATION May 2019

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Thursday, 16th May 2019: 10:00 to 12:00

The paper consists of two sections: A & B.

Section A contains one question, comprising of small parts. [40 marks total]

Section B contains four questions on selected parts of the course. [30 marks each]

Table of Laplace transformations is included at the end of the paper.

Candidates are required to:

*Answer **ALL** parts of Section A and **TWO** questions from Section B.*

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the THREE answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in THREE answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

SECTION A

1. (i) A 100 m long co-axial cable is used to transmit the signal from a sensor. The cable is fabricated with capacitance per metre $C' = 44.3 \text{ pF m}^{-1}$ and inductance per metre $L' = 0.383 \mu\text{H m}^{-1}$.
- Define the characteristic impedance of the cable and calculate its value.
 - Calculate the time duration of the signal passage through the cable.
 - The cable is terminated with a 100Ω resistor. Calculate the amplitude and polarity of the reflected pulse, assuming the initial signal has an amplitude of 20 V.

[9 marks]

- (ii) A signal, which has a finite time duration, has a certain bandwidth in frequency space.
- Explain the relation between the temporal width and frequency bandwidth of the signal, and provide the inequality which their product fulfils.
 - Estimate the duration of a Gaussian pulse of light, using the Full Width at Half Maximum (FWHM) of the light intensity, assuming that the electric field of the pulse is given by

$$E(t) = Ae^{-a^2 t^2}.$$

- (c) Make an estimation for the bandwidth of the same pulse by calculating the FWHM of the power density. **Hint:** You may use the Fourier Transform of the Gaussian pulse:

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$$

which is equal to

$$G(\omega) = e^{-\frac{\omega^2 \sigma^2}{2}}.$$

- (d) Calculate the value of the time-bandwidth product using the FWHM quantities calculated above.

[15 marks]

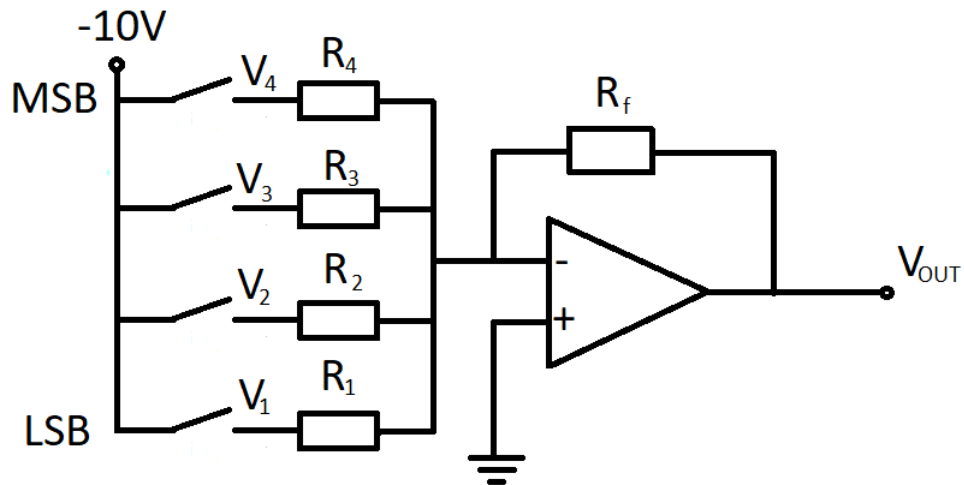
- (iii) A 4-digit Digital to Analog Converter (DAC), shown on the diagram on the following page, has the reference voltage $V_{REF} = -10 \text{ V}$.

- (a) Assuming that the operational amplifier used in the DAC is ideal, using the two 'golden-rules', show that the output voltage V_{out} can be expressed as a function of input voltages V_i and resistances of resistors R_i (where $i=1,2,3,4$) and R_f as follows:

$$V_{out} = -(V_1/R_1 + V_2/R_2 + V_3/R_3 + V_4/R_4)R_f.$$

- (b) Propose values of resistances R_i (where $i=1,2,3,4$), which allow the DAC to work in the output voltage range of 0-15 V, with the Most Significant Bit (MSB) to produce 8 V at the output, assuming that $R_f=800 \Omega$.

[This question continues on the next page ...]



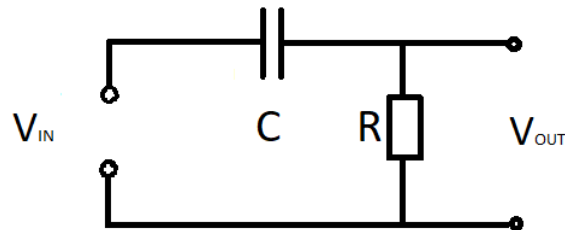
- (c) How many different values of the output voltage can be decoded by this DAC?
- (d) Briefly state the advantages and disadvantages of this type of DAC.

[16 marks]

[Total 40 marks]

SECTION B

2. (i) Consider the RC circuit shown in the diagram below. The input voltage V_{in} is the step function ($V_{in}(t) = 0$ for $t < 0$ and $V_{in} = V$ for $t \geq 0$) and the capacitor is initially charged to a voltage $V/2$. Consider the behaviour of the circuit for $t > 0$.



- (a) Write the differential equation governing the current flowing in the circuit.
- (b) Derive the expression for the current flowing in the circuit as a function of time.
- (c) Calculate and sketch the output voltage V_{out} .

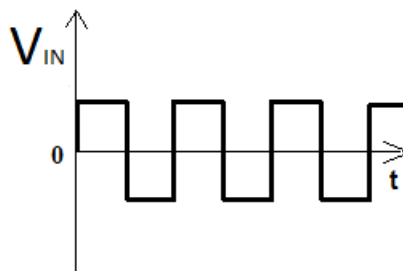
[11 marks]

- (ii) Now consider the same circuit as shown in the diagram above, but assume the input voltage is provided by an AC voltage source.

- (a) Assuming $R=4\text{ k}\Omega$ and $C=0.25\text{ }\mu\text{F}$ calculate the cut-off frequency of the circuit and make a sketch of the Bode plot.
- (b) Explain the behaviour of this circuit as a filter and propose methods to improve its behaviour in terms of steepness of the frequency roll-off, flatness of the pass-band and sharpness of the knee.

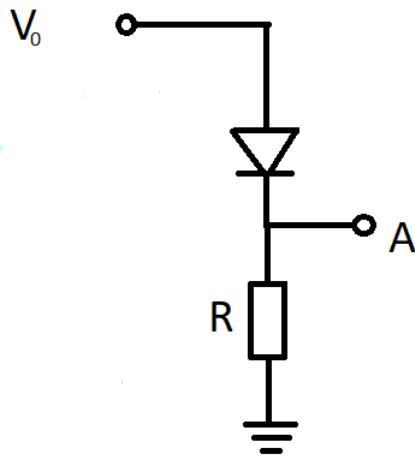
[7 marks]

- (iii) Sketch the output voltage V_{out} of this circuit for the ideal rectangular input signal, shown in the figure below, assuming the input signal has a frequency much smaller than the cut-off frequency of the filter. Comment briefly on its shape.



[4 marks]

- (iv) In the diagram below a voltage V_0 of 2 V is connected to the resistor with resistance $R=100\ \Omega$ via a diode. The voltage drop across the diode is about 0.6 V.



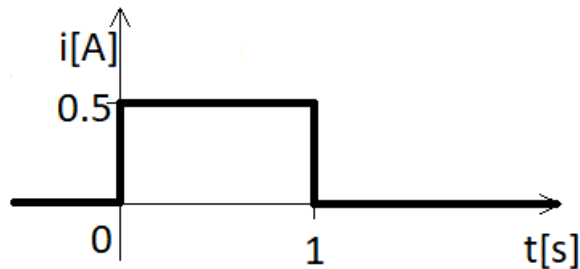
- (a) Calculate the thermal and shot noise contributions to the voltage at point A measured with an ideal voltmeter with a bandwidth of 10 kHz, at temperature of 300 K. What is the total RMS voltage noise from the two sources?
- (b) At which temperature will the shot noise contribution be one order of magnitude larger than the thermal one?

[8 marks]

[Total 30 marks]

3. (i) (a) Define the Laplace transform $F(s)$ of the continuous function $f(t)$.
 (b) Prove that the Laplace transform of the pulse $g(t)$ of unit length and peak current value of 0.5 A, as shown in the diagram below is equal to:

$$\mathcal{L}\{g(t)\} = 0.5(1 - e^{-s})/s.$$



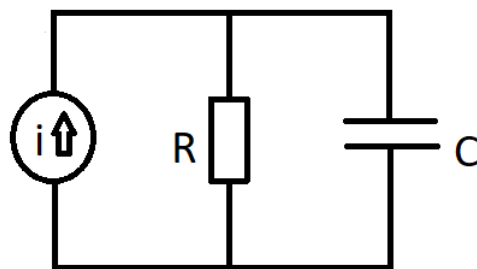
Hint: You may use frequency shifting property of the Laplace transform:

$$\mathcal{L}\{f(t - \lambda)u(t - \lambda)\} = e^{-s\lambda}\mathcal{L}\{f(t)\}$$

where $u(t)$ is the unit step.

[7 marks]

- (ii) The circuit shown below consists of an ideal current source producing the pulse $g(t)$ as shown in the diagram above, a resistor of $2\ \Omega$ and a capacitor of $0.5\ \text{F}$. Assume that, at time $t = 0$, the capacitor is charged and its voltage is equal to 1 V.



- (a) Write the equation describing the conservation of current flowing in the circuit in terms of the partial currents flowing through the resistor and the capacitor, and the pulse produced by the current source.

- (b) Write the differential equation describing the time evolution of the voltage across the capacitor.
- (c) Take the Laplace transform of the equation describing the voltage across the capacitor and write its corresponding s -domain representation.
- (d) By performing the inverse Laplace transform, find an expression for the voltage across the capacitor for time $t > 0$ induced by the current pulse $g(t)$. **Hint:** You may use the table of standard Laplace Transforms at the end of this paper.
- (e) Sketch the voltage across the capacitor for time $t > 0$.

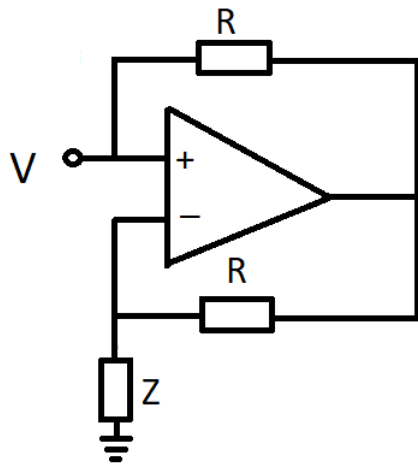
[23 marks]

[Total 30 marks]

4. (i) (a) Explain what is meant by the 'output impedance' of a source.
 (b) Explain the term 'input impedance' as applied to a device.

[4 marks]

- (ii) (a) Briefly state the properties of the ideal operational amplifier and give the two 'golden-rules' used to simplify the analysis of circuits containing them.
 (b) The diagram below shows a circuit with an op-amp which is assumed to be ideal. Show that the input impedance of the device is equal to $-Z$, where Z represents the value of the impedance of the component shown on the diagram.



- (c) Assuming that Z is purely resistive and equals $10\ \Omega$, explain what happens when a positive voltage from 3 V battery is connected to the device. What is the difference when Z is directly connected to the battery?
 (d) Briefly state the properties of typical 'real-world' operational amplifiers.

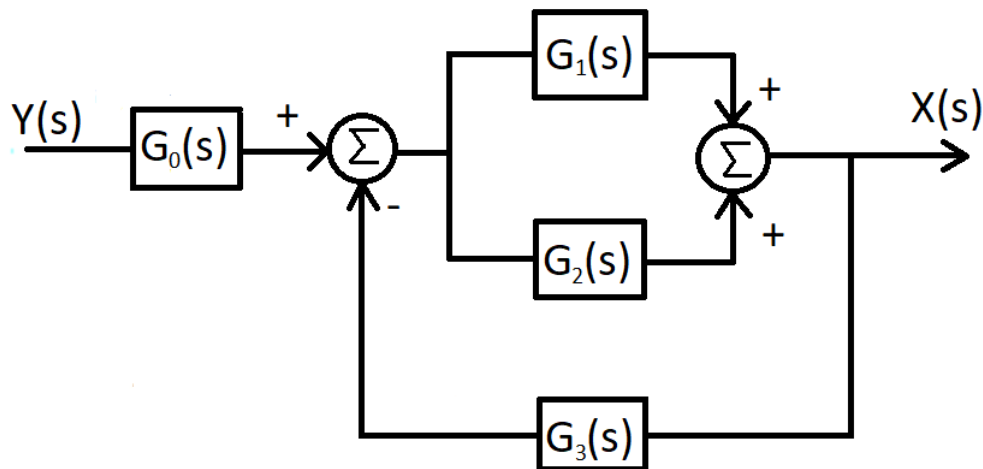
[15 marks]

- (iii) (a) Explain briefly the properties of Flicker noise.
 (b) A resistor with resistance of $112.7\ \Omega$ is operated at room temperature (300 K). Calculate the frequency at which the Flicker noise power density is a factor of two larger than the thermal noise power density, if both contributions are equal at 5 Hz. Calculate power densities for both Flicker and thermal noise at this frequency.
 (c) Calculate the total voltage noise at this frequency, assuming that the measurement is performed with 4 Hz bandwidth.

[11 marks]

[Total 30 marks]

5. The system shown below contains four functional blocks and contains a feedback loop and a parallel connection



- (i)
 - (a) Calculate the system transfer function $G(s)$ in terms of $G_0(s)$, $G_1(s)$, $G_2(s)$ and $G_3(s)$.
 - (b) $G_0(s)$ has a time-domain impulse response of te^{-t} . Find its s -domain representation.
 - (c) $G_1(s)$ is the unity-gain ideal differentiator. Find its s -domain representation.
 - (d) $G_2(s)$ and $G_3(s)$ are ideal amplifiers with gains 2 and 3 respectively. What are their s -domain representations?
 - (e) Find the overall transfer function in terms of s .
- (ii)
 - (a) Find the zeros and poles of the transfer function.
 - (b) Verify if the system is stable and comment on its behaviour in response to a transient input.
 - (c) Determine the system response in the time-domain, to a unit step input.

[15 marks]

[15 marks]

Hint: You may use the table of standard Laplace Transforms at the end of this paper.

[Total 30 marks]

Table of Laplace Transforms

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{-at} \sin(\omega t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos(\omega t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$